

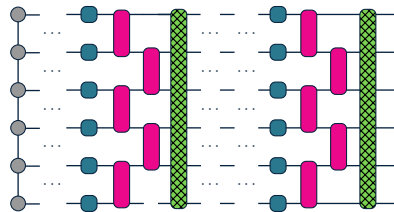
Distributions of Noisy Expectation Values over Sets of Measurement Operators

Matthew Dushenes*, Roger Melko, Juan Carrasquilla, Raymond Laflamme

Institute for Quantum Computing, Perimeter Institute & ETH Zurich

Seminar

Summer, 2026



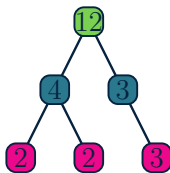
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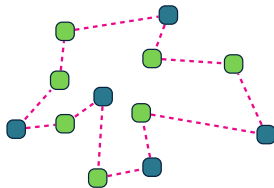
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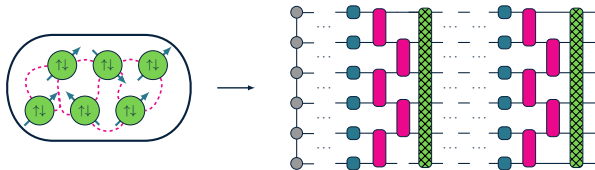
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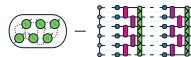
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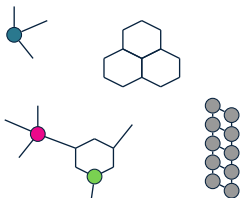
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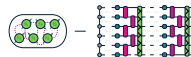
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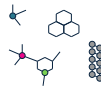
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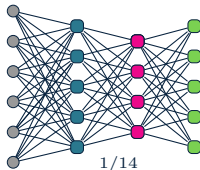
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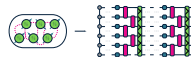
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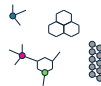
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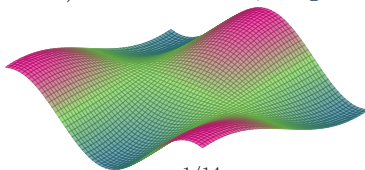
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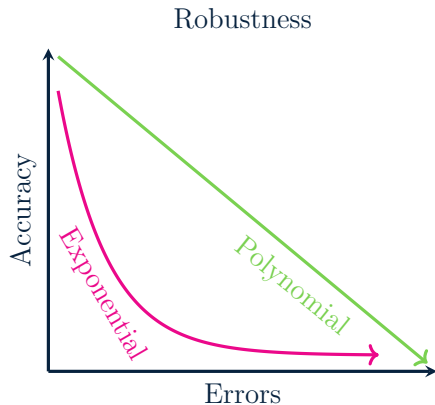
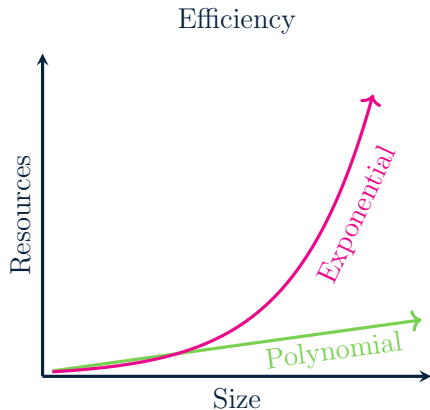


What Makes Problems *Hard*?

- What makes problems *hard* to solve? [1]

What Makes Problems *Hard*?

- What makes problems *hard* to solve?
 - i.e) How do required *resources* increase as problems get *harder*?
 - i.e) How does *accuracy* decrease as problems get *noisier*?

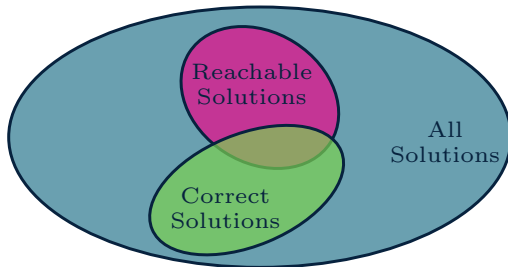


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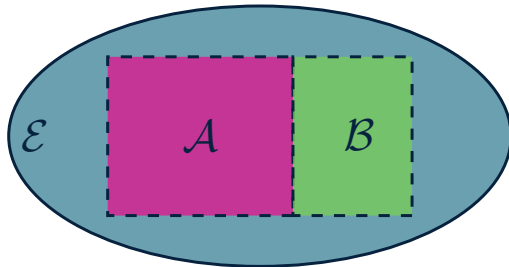
- What makes problems *hard* to solve?
- What does it mean to *solve* a problem?
 - i.e) How close can we get to the *correct* answer if there are *errors*?



Properties of Quantum Systems

Evolutions via channels vs. unitaries are *higher dimensional* and *constrained*

$$\underbrace{U|\psi\rangle \in \mathbb{C}^d}_{\text{Arbitrary Normalized Vector}} \rightarrow \underbrace{\Lambda(\rho) \in \mathbb{C}^{d \times d}}_{\text{Positive-Semidefinite, Hermitian } s\text{-rank Matrix}}$$



Properties of Quantum Systems

Suppose we consider *properties*, or *expectation values* of quantum systems,

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$$\Pi = \sum_{\xi} \xi I_{\xi} \quad , \quad \sum_{\xi} I_{\xi} = I \quad , \quad \sigma \leq \xi \leq \lambda \quad (4)$$

Random States, Operators, and Expectation Values

If states are *randomly distributed* $\rho \sim P_\rho$, given ,

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How are their resulting properties $x \sim P_\Pi$ also *randomly distributed* ? [2, 3, 4]

[2] Zyczkowski, K. *et al.*, *J. Phys. A* **34**, 7111–7125 (2001)

[3] Sauliere, A. *et al.*, *Phys. Rev. B* **112**, 134312 (2025)

[4] Campos Venuti, L. *et al.*, *PLA* **377**, 1854 (2013)

Random States, Operators, and Expectation Values

Suppose states $\rho = \varphi\varphi^\dagger$

are in d -dimensional *systems* and s -dimensional *environments*

with $d \times s$ -dimensional parameters φ , and *positive* operators $\Pi = \Gamma^\dagger\Gamma$.

Random States, Operators, and Expectation Values

Distributions of parameters, states, and expectation values are then related,

$$\varphi \sim P_{\varphi}(\varphi)$$

$$\rho \sim P_{\rho}(\rho) = \int d\varphi \, P_{\varphi}(\varphi) \, \delta(\rho - \varphi\varphi^{\dagger}) \quad (6)$$

$$x \sim P_{\Pi}(x) = \int d\rho \, P_{\rho}(\rho) \, \delta(x - \tau_{\Pi}(\rho)) \, .$$

Random States, Operators, and Expectation Values

Suppose the ds -dimensional *purified* states,

$$\varphi \cong |\psi\rangle \quad \rightarrow \quad \rho = (I_d \otimes \text{tr}_s)(\psi) = \varphi\varphi^\dagger \quad (7)$$

are *uniformly* or *Haar* distributed [2] ,

$$P_\varphi(\varphi)d\varphi \sim \delta(\|\varphi\|^2 - 1) d\varphi \cong d\psi . \quad (8)$$

Random States, Operators, and Expectation Values

From this uniformity and unitary-invariance,

property distributions only depend on *spectra* $\{\xi, d_\xi\}$ of operators Π

and we have the *shift-scale* invariance $\forall \alpha, \beta,$

$$P_\Pi(x) = \frac{1}{\alpha - \beta} P_{\frac{\Pi - \beta I}{\alpha - \beta}} \left(\frac{x - \beta}{\alpha - \beta} \right) \quad (9)$$

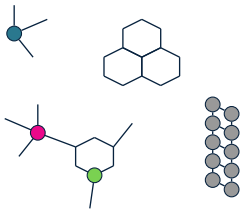
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- *Measurements, Energies, Fidelities* i.e) Hamiltonians [5, 6, 7] $\Pi = \sum_P \pi_P P$



[5] Renes, J. M. *et al.*, *J. Math. Phys.* **45**, 2171–2180 (2003)

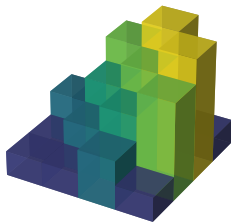
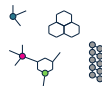
[6] Yashin, V. I. *et al.*, *New J. Phys.* **22**, 103026 (2020)

[7] Meyer, J. J., *Quantum* **5**, 539 (2021)

Expectation Values and Randomness in Quantum Settings

Many *interpretations* and *applications*:

- *Measurements, Energies, Fidelities* i.e) Hamiltonians $\Pi = \sum_P \pi_P P$
- *Tomography* i.e) Classical shadow techniques [8, 9]



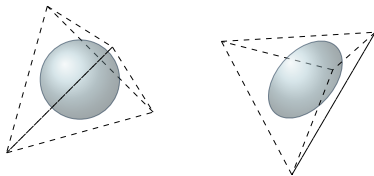
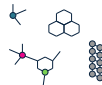
[8] Huang, H.-Y. *et al.*, *Nat. Phys.* **16**, 1050–1057 (2020)

[9] Carrasquilla, J. *et al.*, *Nat. Mach. Intel.* **1**, 155–161 (2019)

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- *Random Algorithms* i.e) Randomized benchmarking, error mitigation [10, 11]

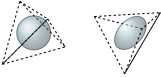


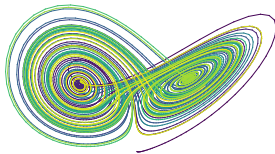
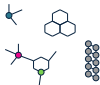
[10] Emerson, J. *et al.*, *J. Opt. B* **7** (2005)

[11] Quek, Y. *et al.*, *Nat. Phys.* **20**, 1648–1658 (2024)

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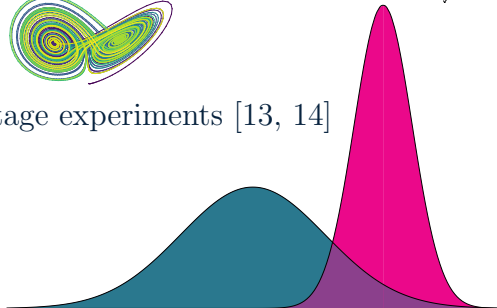
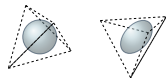
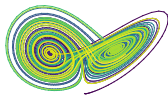
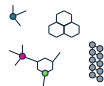
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- *Random Circuit Sampling* i.e) Quantum advantage experiments [13, 14]



[13] Deshpande, A. *et al.*, *PRX Quantum* **3** (2022)

[14] Dalzell, A. M. *et al.*, *Comm. Math. Phys.* **405** (2021)

l -rank projectors $\Pi = I_\Pi$:

$$P_\Pi(x) \sim x^{ls-1} (1-x)^{(d-l)s-1} \quad (10)$$

$$x_t = \frac{\Gamma_{ds}}{\Gamma_{ls}} \frac{\Gamma_{(d-l)s+t}}{\Gamma_{ds+t}} \rightarrow \begin{cases} (l/d)^t & \begin{matrix} d \rightarrow \infty \\ s \rightarrow \infty \end{matrix} \\ t!/d^t & \begin{matrix} d \rightarrow 1 \\ s \rightarrow 1 \end{matrix} \end{cases} \quad (11)$$

Analytical Distributions

#-eigenvalue Hermitian $\Pi = \sum_{\xi} \xi I_{\xi}$:

$$P_{\Pi}(x) = \begin{cases} \sum_{\xi, l_{\xi} \in [d_{\xi}s]} \pi_{l_{\xi}} \operatorname{sign}(\xi - x) (\xi - x)^{(d-d_{\xi})s+l_{\xi}-1} & \# > 1 \\ \frac{1}{\lambda - \sigma} \frac{\Gamma_{ds}}{\Gamma_{d_{\sigma}s} \Gamma_{d_{\lambda}s}} \left(\frac{x - \sigma}{\lambda - \sigma} \right)^{d_{\lambda}s-1} \left(\frac{\lambda - x}{\lambda - \sigma} \right)^{d_{\sigma}s-1} & \# = 2 \\ \delta(x - \lambda) & \# = 1 \end{cases} \quad (12)$$

$$x_t = \sum_{\xi, l_{\xi} \in [d_{\xi}s]} \chi_{l_{\xi},t} \xi^t \quad (13)$$

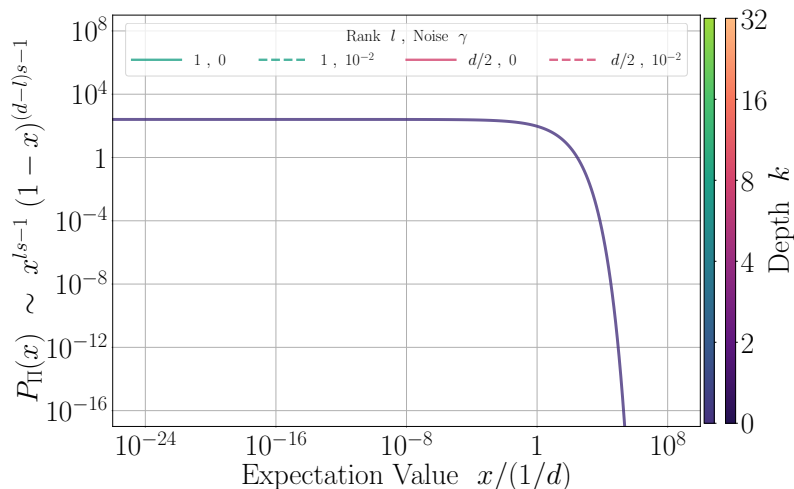
Analytical Distributions

We derive explicit forms for the *moments*, using combinatorics approaches, which are the *complete homogenous symmetric polynomials* in the spectra [15],

$$\pi_{l_\xi} = \frac{1}{2} (-1)^{l_\xi} \frac{\Gamma_{ds}}{\Gamma_{d_\xi s - l_\xi} \Gamma_{(d-d_\xi)s + l_\xi}} \times \sum_{\sum_{\zeta \neq \xi} l_\zeta = l_\xi} \prod_{\zeta \neq \xi} \binom{d_\zeta s + l_\zeta - 1}{l_\zeta} \frac{1}{(\xi - \zeta)^{d_\zeta s + l_\zeta}} \quad (14)$$

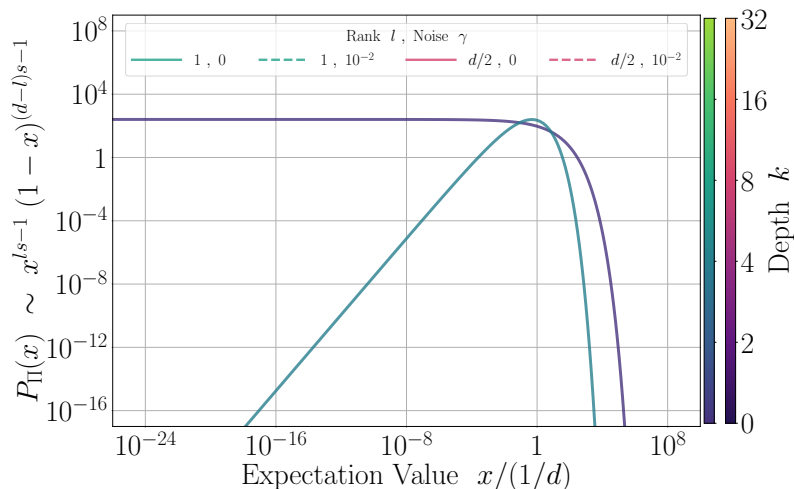
$$\chi_{l_\xi, t} = (-1)^{l_\xi} \frac{\Gamma_{ds}}{\Gamma_{d_\xi s - l_\xi}} \frac{\Gamma_{d_\xi s - l_\xi + t}}{\Gamma_{ds + t}} \times \sum_{\sum_{\zeta \neq \xi} l_\zeta = l_\xi} \prod_{\zeta \neq \xi} \binom{d_\zeta s + l_\zeta - 1}{l_\zeta} \frac{\xi^{d_\zeta s} \zeta^{l_\zeta}}{(\xi - \zeta)^{d_\zeta s + l_\zeta}}. \quad (15)$$

Generalized Porter-Thomas Distributions



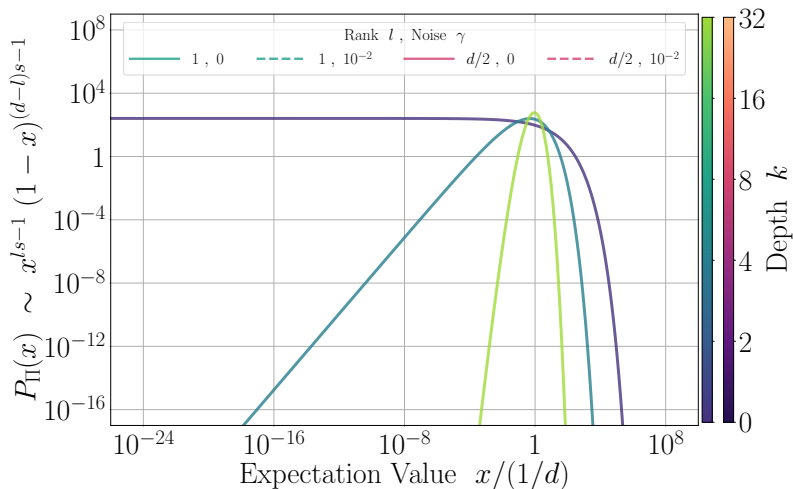
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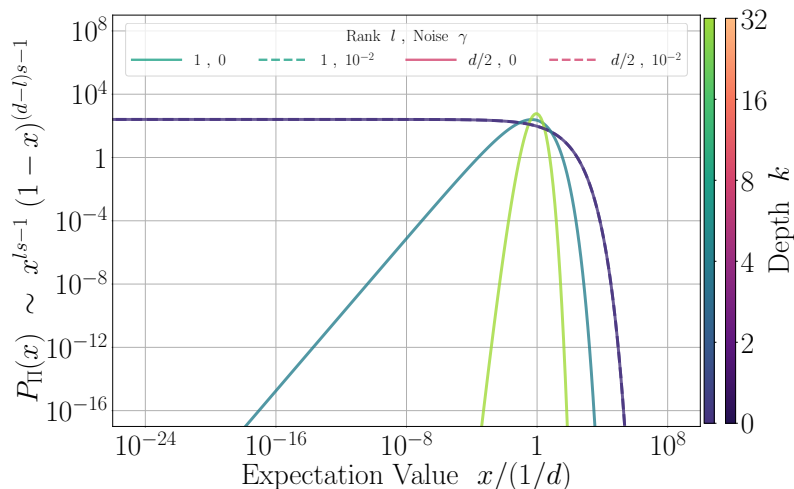
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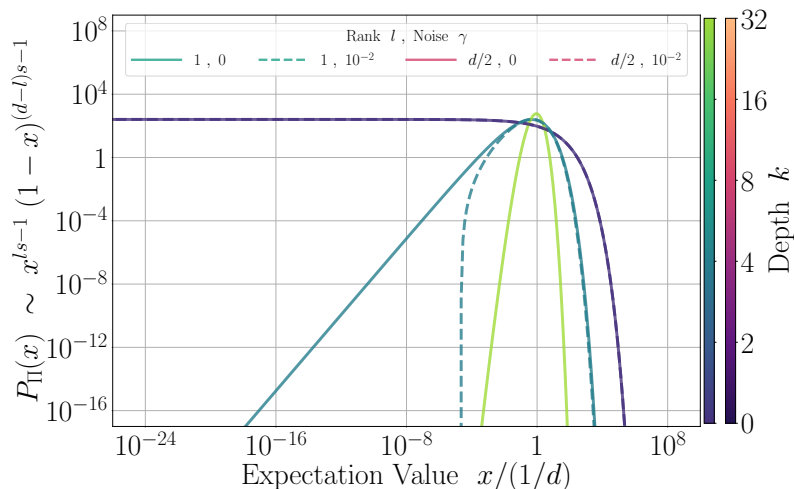
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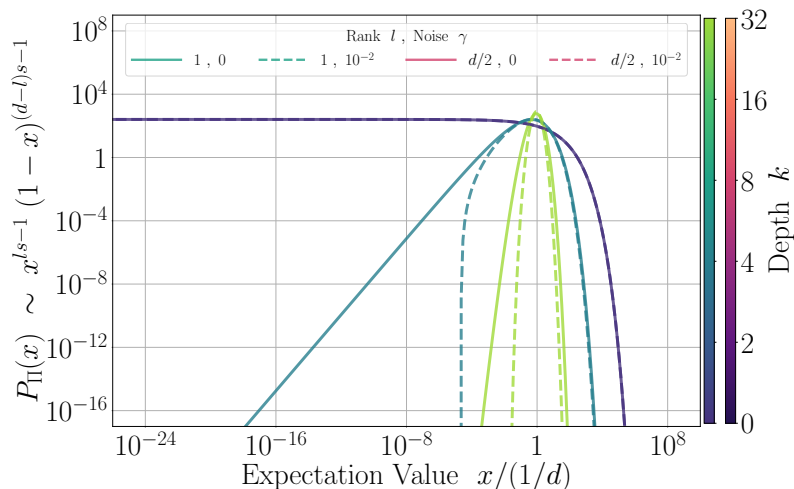
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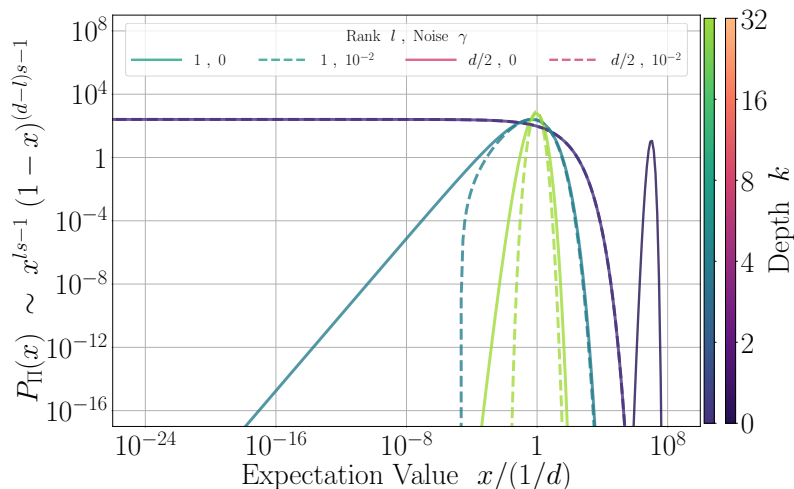
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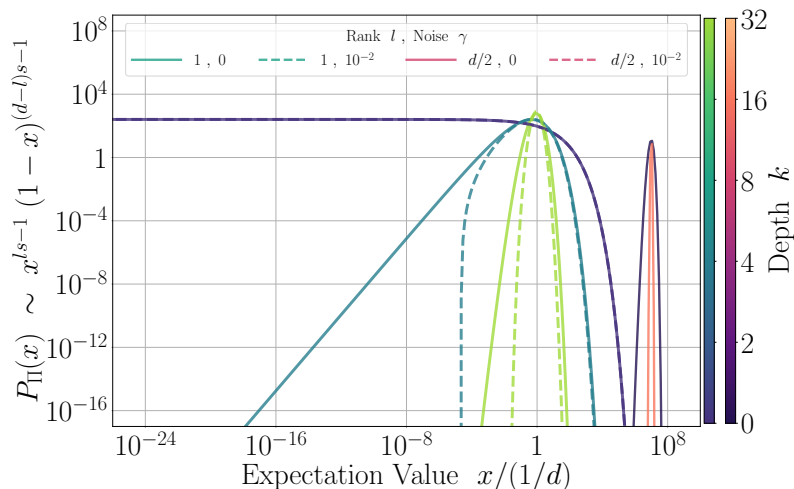
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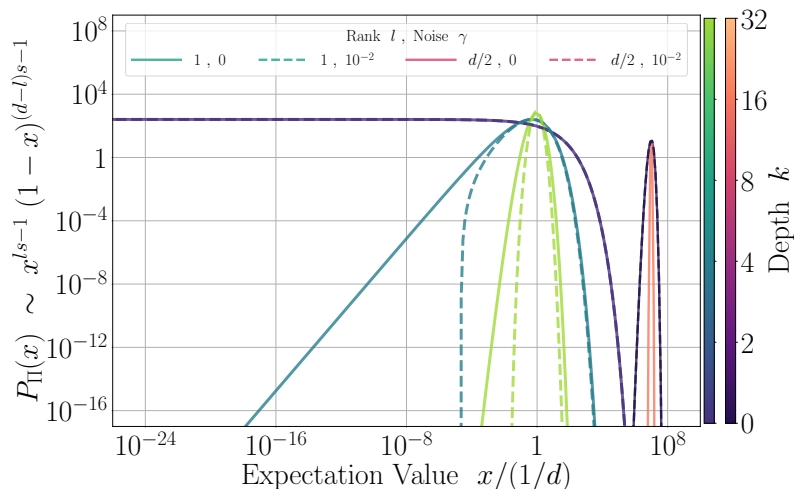
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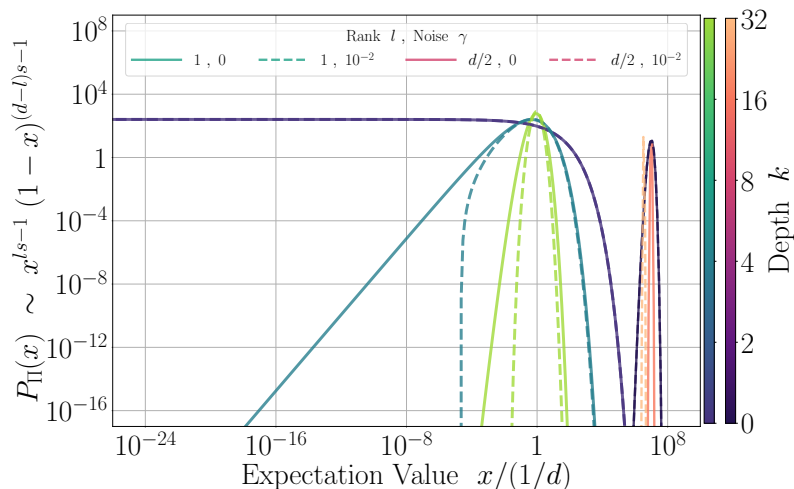
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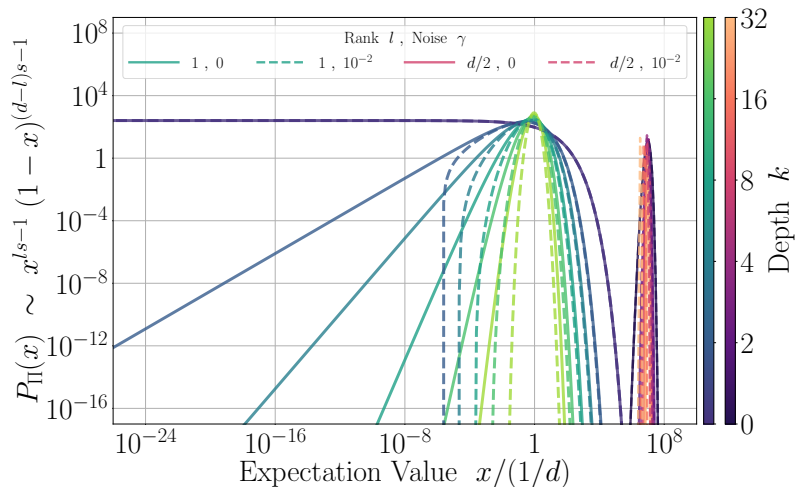
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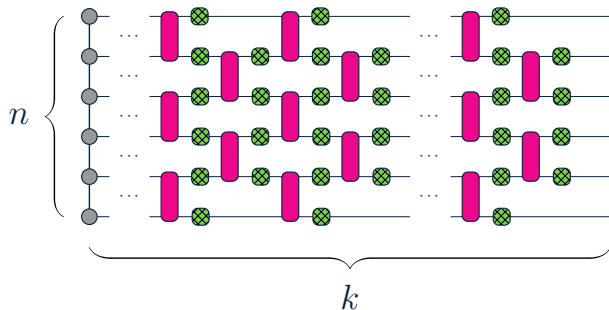
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Noisy Phenomena of Quantum Systems



How do the *noise scale* γ , *circuit depth* k , and *environment size* s affect the *form* of distributions $P_{\Pi}(x)$? [4, 16]

Expectation Values with Noise

Suppose we have a *global* depolarizing noise γ ,

$$\rho \rightarrow \rho_\gamma = (1 - \gamma)\rho + \gamma \frac{\text{tr}(\rho)}{d} I \quad , \quad \Pi \rightarrow \Pi_\gamma = (1 - \gamma)\Pi + \gamma \frac{\text{tr}(\Pi)}{d} I$$

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$\Leftrightarrow$

$$x \rightarrow x_\gamma = (1 - \gamma)x + \gamma \frac{\text{tr}(\Pi)}{d} \quad (16)$$

Expectation Values with Noise

Suppose we have a *global* depolarizing noise γ ,

$$\rho \rightarrow \rho_\gamma = (1 - \gamma)\rho + \gamma \frac{\text{tr}(\rho)}{d} I \quad , \quad \Pi \rightarrow \Pi_\gamma = (1 - \gamma)\Pi + \gamma \frac{\text{tr}(\Pi)}{d} I$$

$$\Leftrightarrow$$

$$x \rightarrow x_\gamma = (1 - \gamma)x + \gamma \frac{\text{tr}(\Pi)}{d} \quad (16)$$

from the *shift-scale* invariance, for *any* noise scale γ , environment dimension s ,

$$P_{\Pi_\gamma}(x_\gamma) = \frac{1}{1 - \gamma} P_\Pi \left(\frac{x_\gamma - \gamma \text{tr}(\Pi)/d}{1 - \gamma} \right)$$

Expectation Values with Sets of Measurements

Given *sets* of measurements, i.e) *informationally complete (IC-POVM)*

$$\mathcal{P} = \{\Pi : \Pi \geq 0\} \quad : \quad \sum_{\Pi \in \mathcal{P}} \Pi = I \quad , \quad \{\tau_{\Pi}(\rho)\}_{\Pi \in \mathcal{P}} \neq \{\tau_{\Pi}(\sigma)\}_{\Pi \in \mathcal{P}} \quad \forall \rho \neq \sigma \quad (17)$$

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Measurement expectation values are *probabilities*,

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PVM (Projector)

$$\Pi = I_{\Pi} \quad (20)$$

$$P_{\mathcal{P}}(p) \sim P_{\Pi}(p) \sim p^{ls-1} (1-p)^{(d-l)s-1}$$

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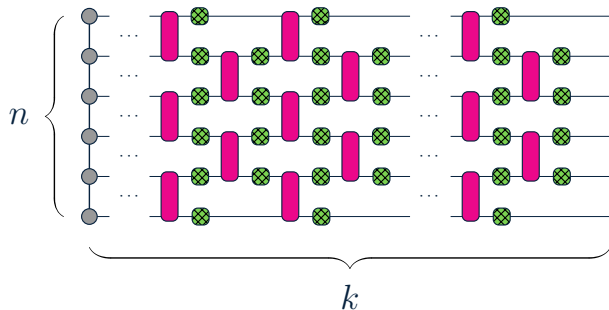
$$P_{\mathcal{P}}(p) \sim P_{\Pi}(p)$$

NON-SIC-POVM (Non-Projector)

$$\Pi = \sum_{\xi} \xi I_{\xi} \quad (22)$$

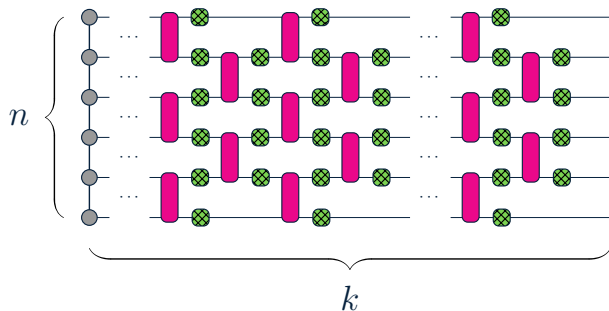
$$P_{\mathcal{P}}(p) \sim \sum_{\Pi \in \mathcal{P}} P_{\Pi|\mathcal{P}}(\Pi) P_{\Pi}(p)$$

Effective Distributions



Given empirical distributions of m samples $p \sim \tilde{P}_{\mathcal{P}_\gamma}^{(k)}(p)$,
How can we model *local* noise effects using a *global* noise model $P_{\mathcal{P}_\gamma}^{(k)}(p)$?

Effective Distributions



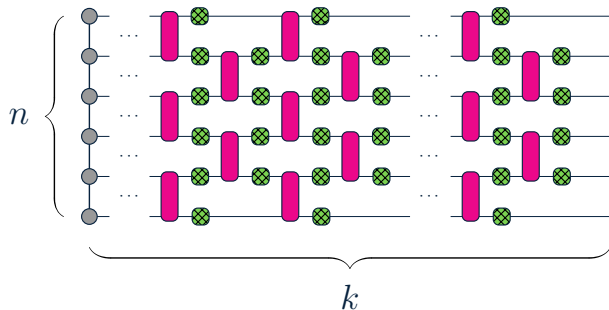
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$$\tilde{P}_{\mathcal{P}_\gamma}^{(k)}(p) \approx P_{\mathcal{P}_\gamma}^{(k)}(p) = \frac{1}{1-\gamma} P_\Pi \left(\frac{x_\gamma - \gamma \text{tr}(\Pi)/d}{1-\gamma} \right) \quad (23)$$

with effective *noise scales* $\tilde{\gamma}$ and *environment dimensions* $\tilde{s}$

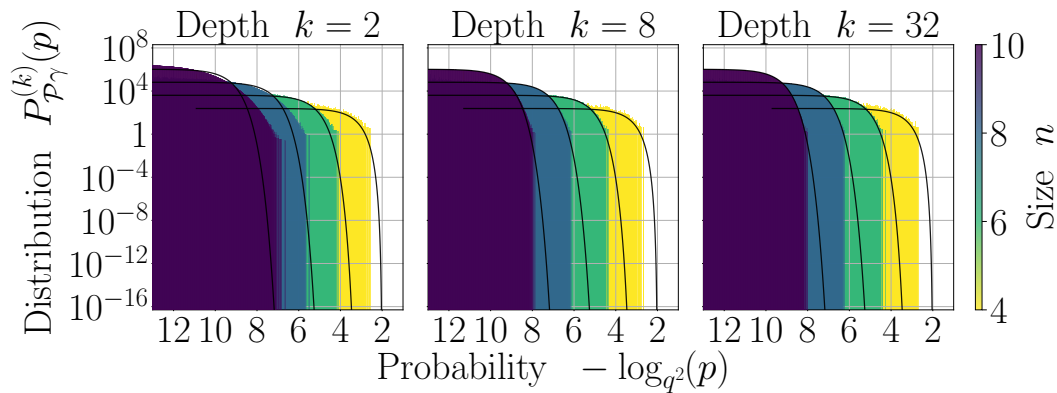
Effective Distributions



In *continuous* variable settings,
Kolmogorov-Smirnov metrics and *binning* are used:

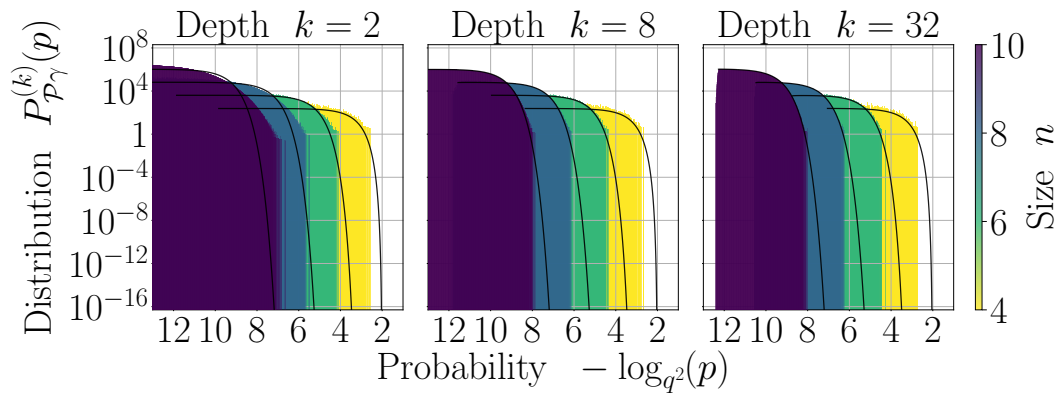
$$\mathcal{L}_{\mathcal{P}_\gamma}^{(k)} = \max_p |F_{\mathcal{F}_\gamma}^{(k)}(p) - \tilde{F}_{\mathcal{P}_\gamma}^{(k)}(p)| \leq \max_{i \in [m]} |F_{\mathcal{F}_\gamma}^{(k)}(p_i) - \tilde{F}_{\mathcal{P}_\gamma}^{(k)}(p_i)| = \tilde{\mathcal{L}}_{\mathcal{P}_\gamma}^{(k)} \quad (24)$$

Empirical Distributions



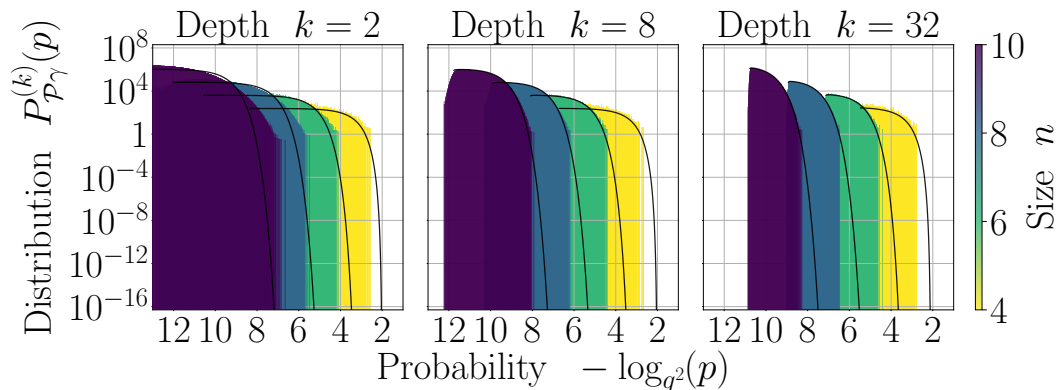
(b) SIC-POVM distributions, for noise scale $\gamma = 0$.

Empirical Distributions



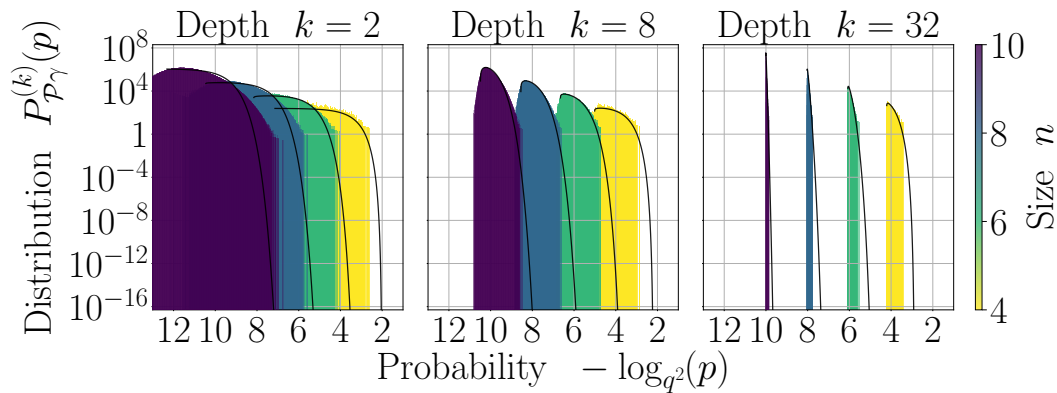
(c) SIC-POVM distributions, for noise scale $\gamma = 10^{-4}$.

Empirical Distributions



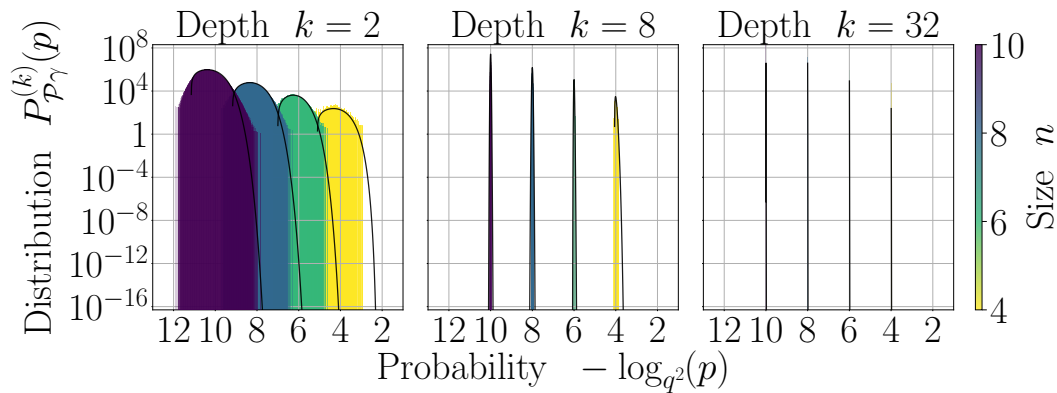
(d) SIC-POVM distributions, for noise scale $\gamma = 10^{-3}$.

Empirical Distributions



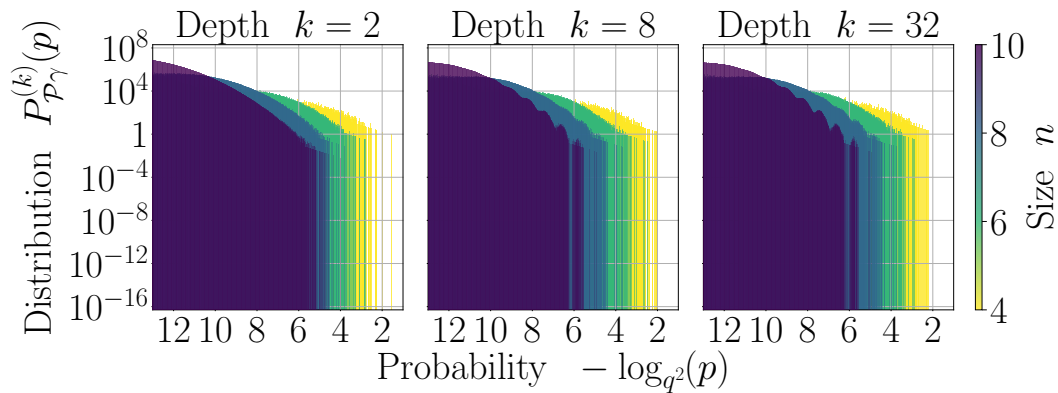
(e) SIC-POVM distributions, for noise scale $\gamma = 10^{-2}$.

Empirical Distributions



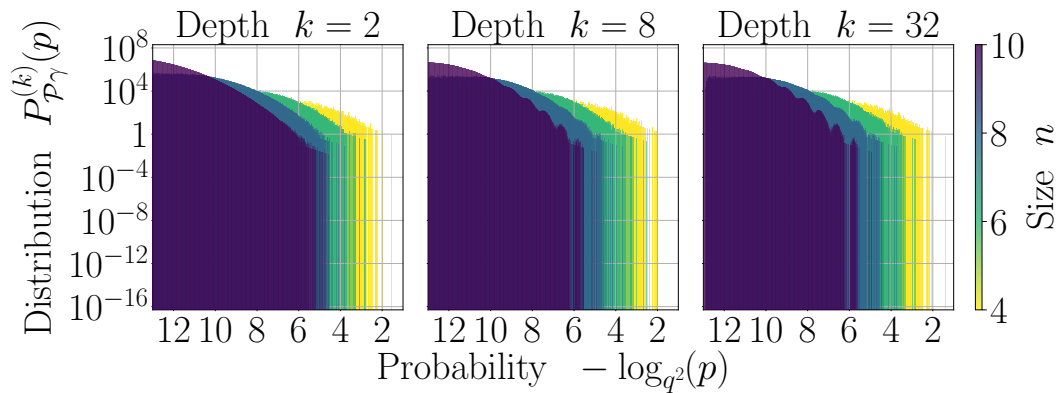
(f) SIC-POVM distributions, for noise scale $\gamma = 10^{-1}$.

Empirical Distributions



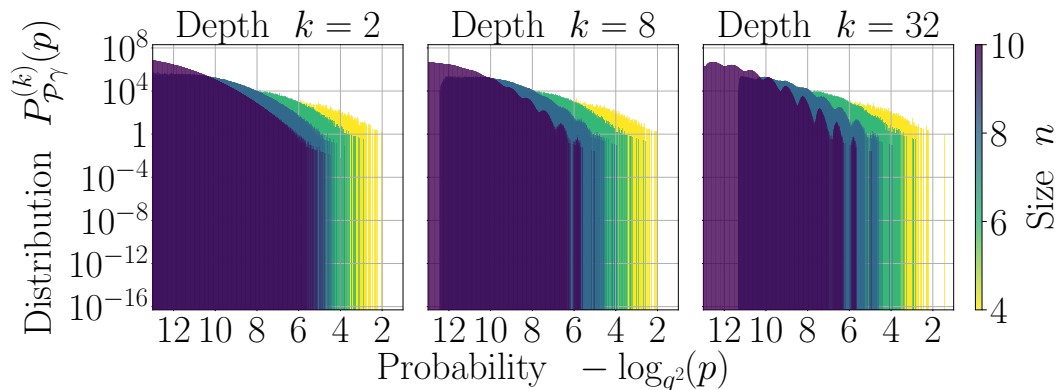
(g) NON-SIC-POVM distributions, for noise scale $\gamma = 0$.

Empirical Distributions



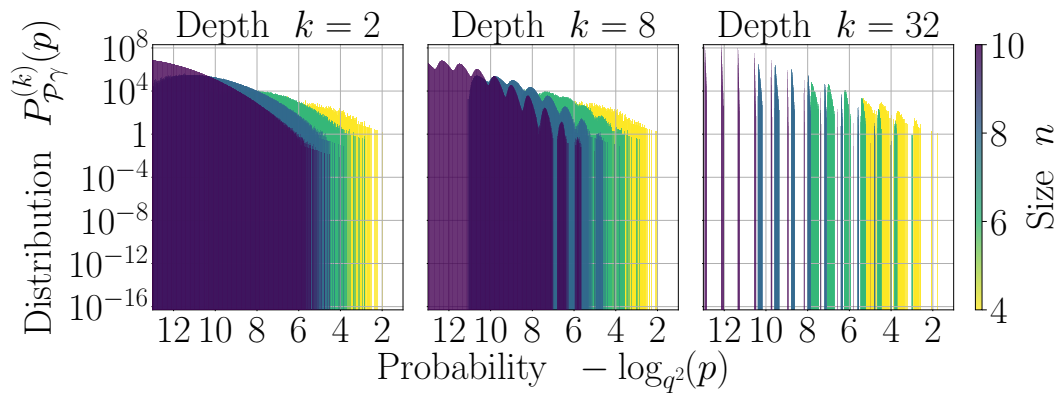
(h) NON-SIC-POVM distributions, for noise scale $\gamma = 10^{-4}$.

Empirical Distributions



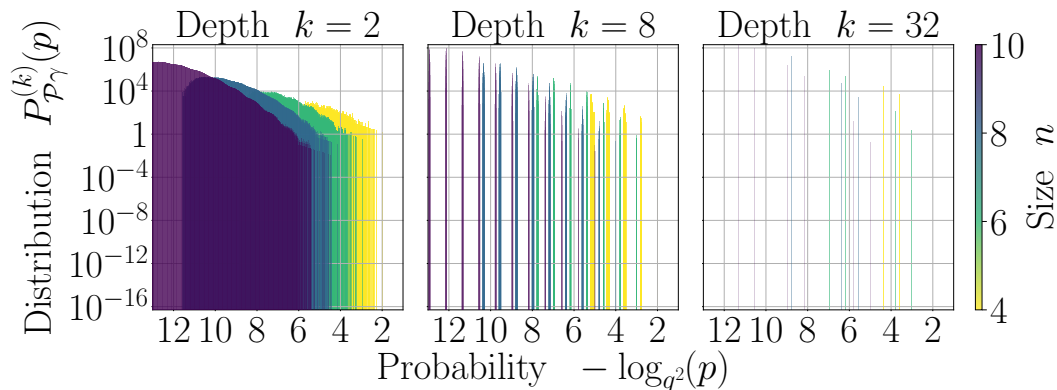
(i) NON-SIC-POVM distributions, for noise scale $\gamma = 10^{-3}$.

Empirical Distributions



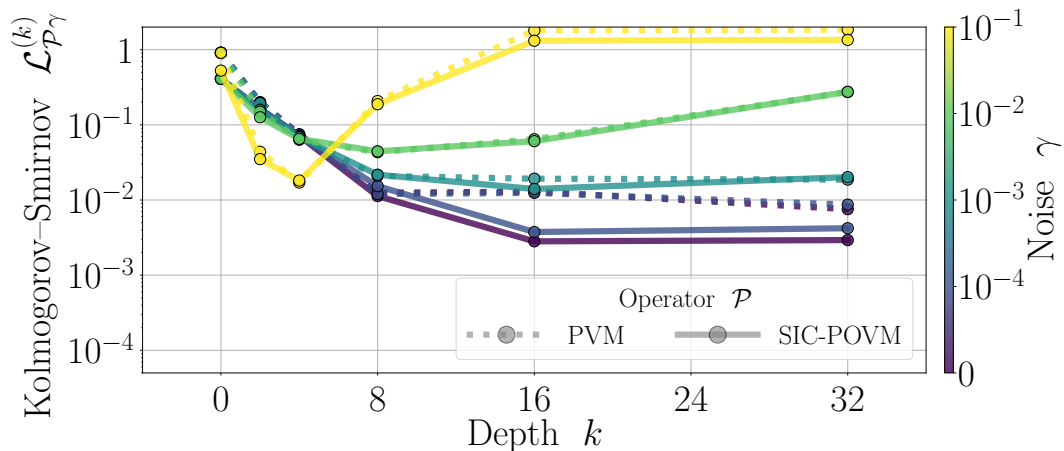
(j) NON-SIC-POVM distributions, for noise scale $\gamma = 10^{-2}$.

Empirical Distributions



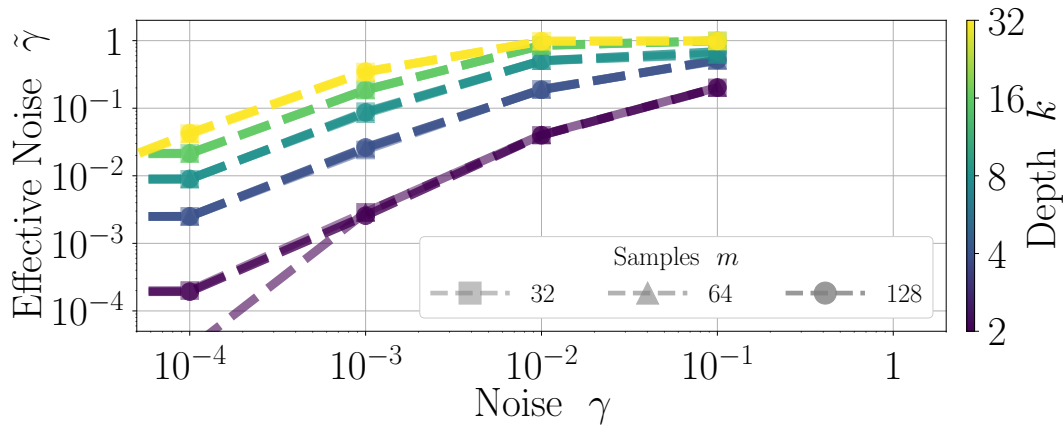
(k) NON-SIC-POVM distributions, for noise scale $\gamma = 10^{-1}$.

Effective Models



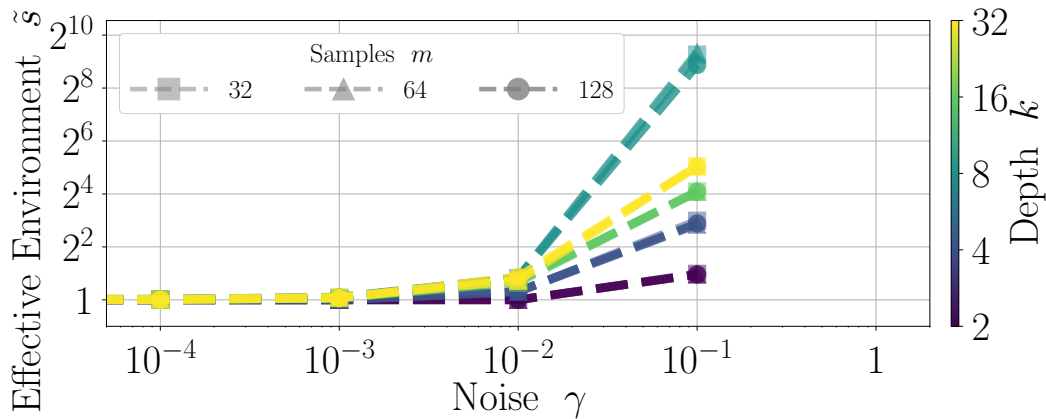
(1) Kolmogorov-Smirnov metrics for system size $n = 10$ and samples $m = 128$.

Effective Models



(m) Effective noise scale for system size $n = 10$.

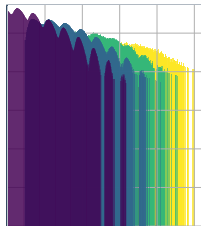
Effective Models



(n) Effective environment dimension for system size $n = 10$.

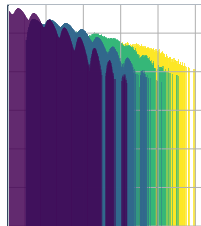
What Have We Learned About Distributions?

- Global, interpretable noise models capture *peaks* behaviours of distributions [16]



What Have We Learned About Distributions?

- Global, interpretable noise models capture *peaks* behaviours of distributions
- Non-symmetric measurements yield *multi-modal* distributions [13, 14]

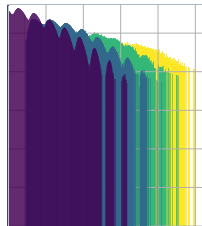


[13] Deshpande, A. *et al.*, *PRX Quantum* **3** (2022)

[14] Dalzell, A. M. *et al.*, *Comm. Math. Phys.* **405** (2021)

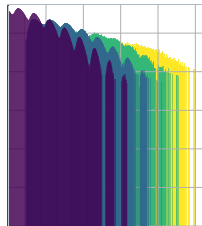
What Have We Learned About Distributions?

- Global, interpretable noise models capture *peaks* behaviours of distributions
- Non-symmetric measurements yield *multi-modal* distributions
- Continuous variables present challenges with determining *sample complexity* [3]



What Have We Learned About Distributions?

- Global, interpretable noise models capture *peaks* behaviours of distributions
- Non-symmetric measurements yield *multi-modal* distributions
- Continuous variables present challenges with determining *sample complexity*
- Which sets of ensembles and operators are *classically simulable*? [17]



Appendix

Moments of Haar Distributions

Moments of Haar-distributed *pure-states* are *symmetric-projectors*,

$$\Psi_t = \int d\psi \, \psi^{\otimes t} = \frac{1}{\binom{d+t-1}{t}} \mathcal{T}_t \quad (25)$$

$$\xi_t = \text{tr}(\mathcal{T}_t \Pi^{\otimes t}) = \sum_{\sum_{\xi} \sum_{l_{\xi} \in [d_{\xi}]} t_{l_{\xi}} = t} \prod_{\substack{\xi \\ l_{\xi} \in [d_{\xi}]}} \xi^{t_{l_{\xi}}} \rightarrow \sum_t \xi_t x^t \quad (26)$$

from *generating functions*, are *complete homogenous symmetric polynomials*

$$F(x) = \sum_t \frac{1}{t!} f_t x^t, \quad G(x) = \sum_t \frac{1}{t} g_t x^t \quad (27)$$

$$F(G(x)) = \sum_t \sum_l \sum_{\substack{\sum_k l_k = l \\ \sum_k l_k k = t}} \frac{1}{l!} \binom{l}{\{l_k\}} f_l \prod_k \left(\frac{g_k}{k}\right)^{l_k} x^t \quad (28)$$

Derivations of Haar Distributions

Given minimal assumptions of *trace-preservation* or *unitary-invariance*,

$$P_{\Pi}(x) = \frac{1}{\frac{1}{2}\Omega_{2ds}} \int d\varphi \delta(\|\varphi\|^2 - 1) \delta(\|\varphi\|^2 - x) \quad (29)$$

which allows *bi-partitioning* $\varphi \rightarrow \varphi \oplus \bar{\varphi}$ to satisfy Dirac-delta constraints,

$$P_{\Pi}(x) \propto \frac{1}{|\det \Gamma|^{2s}} x^{ls-1} \frac{1}{\frac{1}{2}\Omega_{2ls}} \int d\varphi \delta(\|\varphi\|^2 - 1) (1 - x\|\Gamma^{-1}\varphi\|^2)^{(d-l)s-1} \quad (30)$$

with *partial-fraction decomposition* for moments.

Otherwise *contour-integration* techniques can be used [4] .

Continuous Variable Distributions

m independent and identically distributed *samples* $\{x_i \sim P\}_{i \in [m]}$ yield,

$$\tilde{F}(x) = \frac{1}{m} \sum_{i \in [m]} \delta(x \geq x_i) \approx F(x) = \int^x dz P(z) , \quad (31)$$

with means and variances in terms of the distribution F ,

$$\mu_{\tilde{F}}(x) = F(x) \quad , \quad \Sigma_{\tilde{F}}(x) = \frac{1}{m} F(x)(1 - F(x)) . \quad (32)$$

From Chebyshev's inequality, the *sample-complexity* is,

$$P\left(|\tilde{F} - \mu_{\tilde{F}}| \geq \epsilon\right) \leq \frac{\Sigma_{\tilde{F}}}{\epsilon^2} \leq \delta^2 \quad \rightarrow \quad m \geq \left(\frac{1}{2\delta\epsilon}\right)^2 . \quad (33)$$

Continuous Variable Distributions

Let us bin m samples into m' discretized *bins*,

$$\{x_i\}_{i \in [m]} \rightarrow \{x'_i\}_{i \in [m']} \quad : \quad x_i \rightarrow x'_{i'} \text{ if } x_i \in [x'_{i'}, x'_{i'+1}] , \quad (34)$$

with bin density, bin size, and cumulative bin density,

$$\rho(x_{i'}) = \frac{|\{x_i : x_i \in [x'_{i'}, x'_{i'+1}]\}_{i \in [m]}|}{m} \quad (35)$$

$$\eta(x'_{i'}) = \frac{x'_{i'+1} - x'_{i'}}{m'} \quad (36)$$

$$\omega(x'_{i'}) = \sum_{j' \leq i'} \rho(x_{j'}) . \quad (37)$$

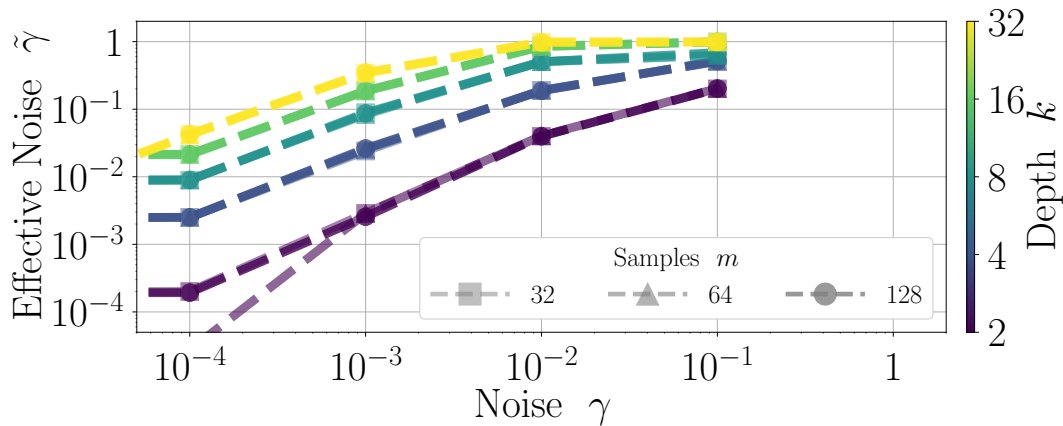
Continuous Variable Distributions

Such $m \rightarrow m'$ binning yields *empirical* distributions,

$$\tilde{P}'(x) = \frac{1}{m'} \sum_{i' \in [m']} \rho(x'_{i'}) \frac{1}{\eta(x'_{i'})} \delta(x'_{i'} \leq x < x'_{i'+1}) \quad (38)$$

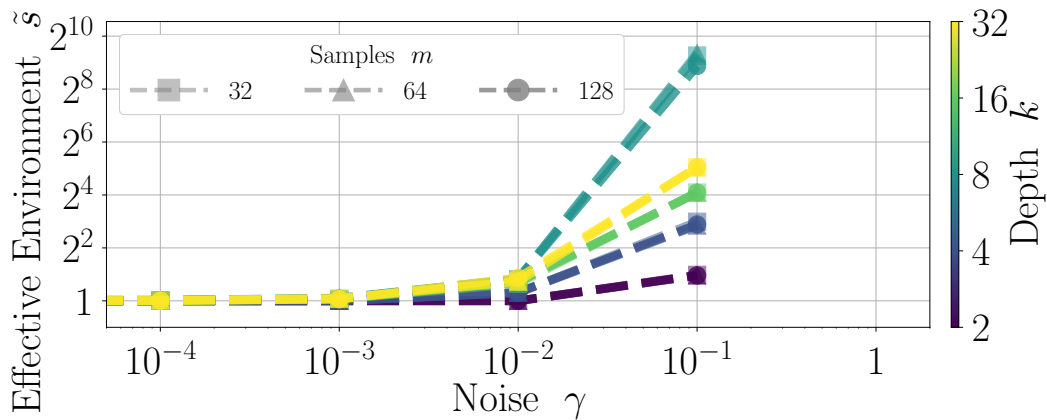
$$\tilde{F}'(x) = \sum_{i' \in [m']} \omega(x'_{i'}) \delta(x \geq x'_{i'}) , \quad (39)$$

Effective Models



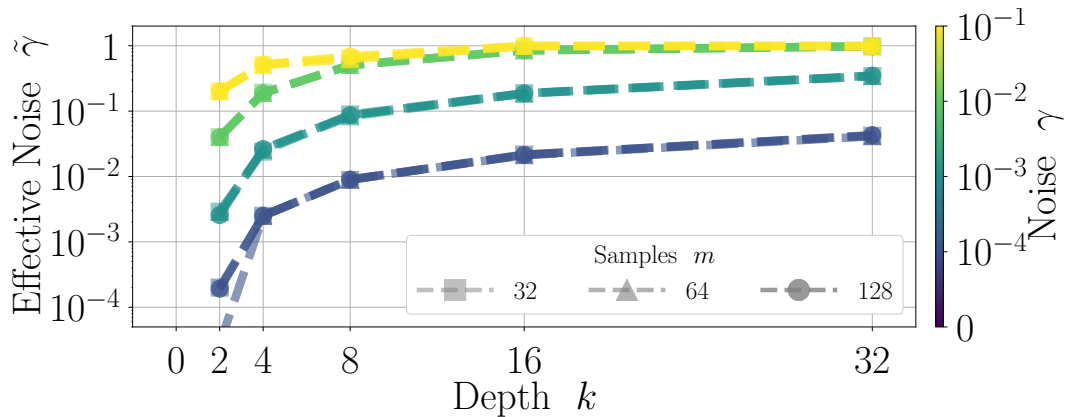
(o) Effective noise scale for system size $n = 10$.

Effective Models



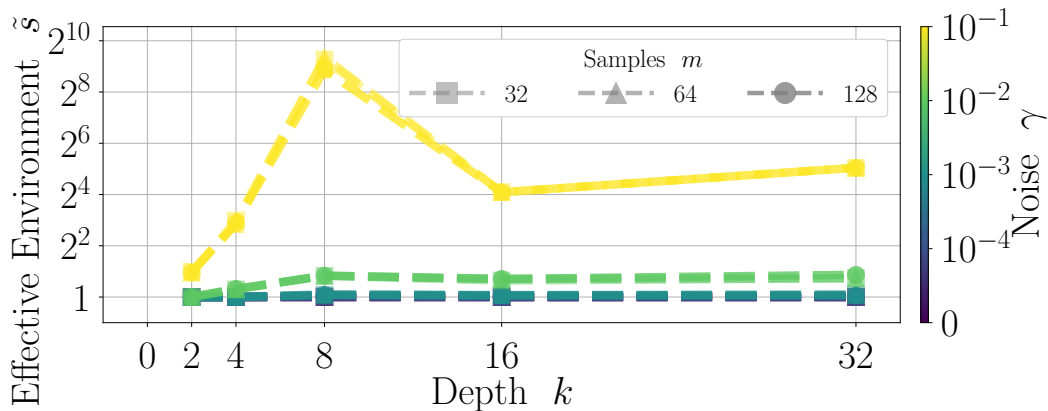
(p) Effective environment dimension for system size $n = 10$.

Effective Models



(q) Effective noise scale for system size $n = 10$.

Effective Models



(r) Effective environment dimension for system size $n = 10$.